Prakul Sherikar

Austin, Texas | 512-704-2855 | Prakulsherikar@utexas.edu | www.linkedin.com/in/prakul-sherikar

EDUCATION

The University of Texas at Austin ***May 2028***
Bachelor of Science in Mechanical Engineering *GPA: 3.74/4.00*

- Activities: IEEE RAS, UT ASME, UT SEEK
- Relevant Coursework: Combustion Engine Processes

EXPERIENCE

Longhorn Baja SAE | Unsprung Lead ***September 2025 - Present***

- Cut peak upright stress 42% (470 to 274 MPa) by investigating a stress concentration from an earlier revision, confirming it under mesh refinement in Ansys, and reverting to a uniform cross-section.
- Delivered both CNC-machined front uprights, which ran at Baja SAE Ohio with no cracks found afterward, designed for manufacturability and assembly with bolt and hand-tool access and 5 mm minimum wheel clearance verified through tolerance analysis within the wheel envelope.
- Caught assembly issues before machining by 3D printing both front uprights and assembling them with bolts, misalignment spacers and the tie rod: enlarged the nut pocket after the nut proved hard to install, resized the tie rod arm hole, and confirmed the weld-cup clearance cutout for the control arms (sized in CAD) worked on the car with no further material removal.
- Defined a 7.15g vertical, 4g lateral and 2g braking load case (above the team's 5 to 6g baseline) from weight-transfer hand calculations benchmarked against other Baja teams, then validated worst cases in Ansys FEA.
- Reduced unsprung mass 15% (2.16 to 1.84 lb) through targeted pocketing and material selection, with suspension kinematics unchanged.
- Verified the hub bearing for steering-downhill and 6 ft drop loads (about 6g) with static, dynamic and L10 life calculations.
- Designed the welded steel backup upright as sole designer and am advising two teammates on this year's CNC uprights.

Independent Project | Liquid Cold Plate Thermal Design Optimization ***June 2026 - Present***

- Cut thermal resistance 21% (0.1275 to 0.1009 C/W) in a parametric Ansys Fluent study of 5 flow rates, 3 channel counts and 2 manifolds on a 150 W, 9.4 W/cm2 heat source.
- Identified pin-fin as the best geometry, with 25% lower thermal resistance than optimized straight channels for only +1.7% pressure drop (serpentine reached 22% lower at 5x the pressure drop).
- Selected 0.75 L/min and 6 channels on diminishing returns (1.25 L/min gained another 24% but raised pressure drop 2.3x), and cut pressure drop 11% (63.3 to 56.3 Pa) with an 8-degree diffuser manifold; CFD contours exposed outer-channel flow maldistribution.
- Verified CFD results with an energy balance closing within 0.2% (149.7 W vs. 150 W) using mass-weighted outlet temperature, and wrote a Python script to compute thermal resistance from the CFD outputs.

IEEE Robotics and Automation Society – Chessbot ***January 2025 - May 2025***

- Cut per-move robotic arm cycle time in half (4 sec to 2 sec), by redesigning the robot's gantry system from a two-bar configuration to CoreXY to remove motor mass from the moving assembly
- Improved ease of piece placement and removal by prototyping and iterating the electromagnet cartridge through 3 revisions to resolve bolt/nut clearance issues
- Demoed the robot at a UT robotics showcase and a Girl Day outreach event

LEADERSHIP & INVOLVEMENT

Longhorn Sim Racing | Competition Officer & Driver ***May 2026 - Present***

- Scaled the "Lone Star Cup" racing league from a single-campus grid into a multi-collegiate league, recruiting drivers from Clemson and Texas State
- Modeled optimal endurance race strategy by logging lap time, fuel consumption, and tire degradation across push/fuel-save stints, then calculating target pace and pit-stop timing

SCCA F&C Marshal | FIA/COTA ***2025 - Present***

- Served as Flagging & Communication Marshal for elite international motorsports series including FIA WEC '25, '26 (Lone Star Le Mans) and SVRA Trans Am

TECHNICAL SKILLS

- **Chassis & Suspension:** Uprights, deriving load cases, bearing life
- **CAD & Modeling:** SolidWorks, Fusion 360, Engineering Drawings, DFM, DFA
- **Simulation & Analysis:** Ansys FEA, Ansys Fluent, CFD, telemetry/data analysis, Thermal Analysis
- **Manufacturing & Testing:** Material yield/Failure analysis, 3D printing, rapid prototyping, welding prep and fixturing/jigging, GD&T fundamentals, Sheet Metal Design, Tolerance Analysis, Packaging
- **Documentation:** BOM management, technical documentation, cross-team design review,
- **Programming:** MATLAB (Coursework), Python

ADDITIONAL INFORMATION

- Interests: Autonomous Robotics, Aerodynamics, Motorsports, and eating/drinking anything sweet (Pizookies & boba)